Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 26)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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6	
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8	
9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 For each of the following investigations, name the most appropriate statistical test.

[4 marks]

Whether noise level decreases as distance from a road increases .....

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Whether the numbers of dormice differ between woodlands with different management ..

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Whether mean dissolved oxygen differs between two streams .....

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Whether crop yield increases with fertiliser application (ranked data) .....

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0 1 . 2 Explain why the chi-squared test cannot be used for data such as wind speed or plant height.

[1 mark]

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0 1 . 3 A Spearman's rank test on 12 pairs of data gave $r_s = -0.72$. The critical value at $p = 0.05$ for $n = 12$ is 0.59. Explain what can be concluded.

[3 marks]

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0 1 . 4 Explain the purpose of error bars on a graph of mean values.

[2 marks]

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0 2 Table 1 shows information about some greenhouse gases.

Table 1

Gas	Main anthropogenic sources	Approximate residence time	Global warming potential (per molecule, relative to CO_2 over 100 years)
Carbon dioxide	Fossil fuel combustion, deforestation, cement	100+ years (variable)	1
Methane	Rice paddies, cattle, landfill, gas leaks, coal mines	About 12 years	28
Nitrous oxide (NOx)	Fertilisers, combustion	About 120 years	265
CFCs	Refrigerants, aerosols (now banned)	50-100+ years	5 000-10 000
Tropospheric ozone	Secondary pollutant from NOx and hydrocarbons	Days-weeks	-

0 2 . 1 Explain the natural greenhouse effect.

[3 marks]

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0 2 . 2 Use Table 1 to explain why carbon dioxide is considered the most important anthropogenic greenhouse gas even though methane and CFCs absorb far more infrared per molecule.

[3 marks]

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0 2 . 3 Explain why reducing methane emissions would produce a faster reduction in warming than reducing CO_2 emissions by the same mass.

[2 marks]

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0 2 . 4 Give **two** methods of reducing methane emissions from agriculture or waste.

[2 marks]

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2 .....

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0 3 A student measured the mass of carbon stored per hectare in three land uses in the same region.

Table 1

Land use	Carbon in vegetation / t ha^{-1}	Carbon in soil dead organic matter / t ha^{-1}
Mature woodland	120	95
Permanent pasture	3	80
Arable (ploughed annually)	2	45

0 3 . 1 Calculate the total carbon lost per hectare when mature woodland is converted to arable farmland.

[1 mark]

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Answer [ ] t/ha

0 3 . 2 Explain why the soil carbon in arable land is so much lower than in pasture, even though the vegetation carbon is similar.

[2 marks]

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0 3 . 3 Suggest **two** ways the arable farmer could increase soil carbon storage.

[2 marks]

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2 .....

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0 4 . 1 Which of the following best describes the **precautionary principle**? Tick **one** box.

[1 mark]

A A waste is assumed to cause pollution until research shows it is unlikely to do so ☒

B The polluter must pay for any damage caused ☐

C Emissions should be as low as reasonably achievable ☐

D Only pollutants known to be toxic need to be controlled ☐

0 4 . 2 Explain why the polluter pays principle works best when penalties are high and enforcement is reliable.

[2 marks]

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0 4 . 3 Give **two** choices about emission location that reduce the severity of pollution.

[2 marks]

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0 5 In 2020 the tanker MV Wakashio ran aground on a coral reef off Mauritius and leaked about 1000 tonnes of fuel oil into a shallow lagoon near mangroves and a marine park.

0 5 . 1 Explain why inflatable booms were deployed immediately, and give **one** limitation of booms.

[3 marks]

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0 5 . 2 Describe how skimmers and absorbent materials remove oil, and state how the collected oil is disposed of.

[3 marks]

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0 5 . 3 Explain why dispersants were **not** used in the lagoon.

[2 marks]

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0 5 . 4 Explain how bioremediation could help remove the oil that remained on the mangrove sediments, and suggest how it could be accelerated.

[2 marks]

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0 6 Table 1 shows the mean surface temperature of the Earth with and without an atmosphere containing greenhouse gases.

Table 1

Condition	Mean surface temperature / °C
No greenhouse effect	-18
Natural greenhouse effect	15

0 6 . 1 Use Table 1 to calculate the warming produced by the natural greenhouse effect.

[1 mark]

Answer ☐ °C

0 6 . 2 Explain how the natural greenhouse effect warms the Earth's surface.

[4 marks]

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0 6 . 3 Explain what is meant by a **greenhouse gas**.

[1 mark]

0 6 . 4 The wavelengths of radiation arriving at the Earth are different from those leaving. Describe this difference and explain why it matters to the energy budget.

[2 marks]

0 6 . 5 Explain why a change in the movement of energy through the atmosphere could affect the distribution of species.

[2 marks]

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0 7 Table 1 shows the volume of water in some reservoirs of the hydrological cycle and the annual transfer rates through them.

Table 1

Reservoir	Volume / x 10^{15} dm^3	Annual transfer (inflow = outflow) / x 10^{15} dm^3 yr^{-1}	Percentage of global water / %
Oceans	1 340 000	425	97
Ice caps and glaciers	24 000	3	1.7
Groundwater	23 000	12	1.7
Lakes and rivers	180	40	0.013
Soil moisture	17	80	0.001
Atmosphere	13	480	0.001

0 7 . 1 The residence time of water in a reservoir is given by: residence time = volume in reservoir / annual transfer rate. Use Table 1 to calculate the residence time of water in the atmosphere. Give your answer to the nearest whole day.

[3 marks]

Answer _____ days

0 7 . 2 Calculate the residence time of water in the oceans.

[1 mark]

Answer _____ years

0 7 . 3 Explain why the atmosphere is an important reservoir even though it holds only 0.001% of global water.

[2 marks]

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0 7 . 4 Explain why human changes to a small reservoir such as soil moisture are more obvious than changes to the oceans.

[2 marks]

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0 7 . 5 Explain why the hydrological cycle is described as being in dynamic equilibrium.

[2 marks]

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0 8 . 1 Explain how each of the following human activities increases soil erosion.

[4 marks]

Overgrazing .....

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Reducing soil biota .....

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0 8 . 2 Explain why landslides become more likely on steep slopes some years after deforestation.

[2 marks]

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0 8 . 3 Describe **four** environmental impacts of accelerated soil erosion away from the field where it occurs.

[4 marks]

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0 9 . 1 Explain the difference between a **mutagen**, a **carcinogen** and a **teratogen**.

[3 marks]

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0 9 . 2 Explain why a birth abnormality caused by a teratogen cannot be inherited by future generations, whereas one caused by a mutagen may be.

[2 marks]

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0 9 . 3 Distinguish between **somatic** and **gonadic** effects of a mutagen.

[2 marks]

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0 9 . 4 Explain what is meant by **synergism** and give an example.

[3 marks]

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1 0 Table 1 shows the mean daily solar energy received on a horizontal surface at three locations.

Table 1

Location	Latitude	Mean solar energy in June / kWh m^{-2} day^{-1}	Mean solar energy in December / kWh m^{-2} day^{-1}
Sahara desert	25°N	7.8	4.2
Southern England	51°N	5.0	0.7
Northern Norway	69°N	4.6	0.0

1 0 . 1 Use Table 1 to calculate how many times more solar energy is received in December in the Sahara than in southern England.

[1 mark]

Answer [ ] times

1 0 . 2 Explain why northern Norway receives more solar energy in June than in December.

[2 marks]

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1 0 . 3 Explain why concentrating solar power systems using parabolic reflectors only work where there is no cloud.

[2 marks]

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1 0 . 4 Explain why the low energy density of solar power is a problem for large-scale generation.

[1 mark]

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1 0 . 5 Evaluate the extent to which new technologies could overcome the problems of using solar power as a large-scale energy resource.

[9 marks]

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Write an essay on one of the following topics.

1 1 . 1 Evaluate the contribution of new technologies to the monitoring and understanding of environmental change in the physical environment.

[25 marks]

OR

1 1 . 2 Mineral resources are non-renewable. Discuss how an understanding of geological processes and the development of new technologies can extend the period for which society can rely on mineral resources.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS